

ECON 712: Applied Econometrics

Lecture 20: Instrumental Variables, Part V — LATE

From Constant to Heterogeneous Treatment Effects

Compliance Types and LATE Assumptions

The LATE Theorem

Interpreting LATE

LATE and 2SLS

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Summary

Limitations of the Constant-Effects Model

- So far we assumed: $y_i = \alpha + \rho D_i + \eta_i$, where ρ is the **same** for everyone
- This rules out that individuals respond differently to treatment — a strong assumption
- In reality, treatment effects are **heterogeneous**:
 - ▶ Education raises wages more for some people than others
 - ▶ Military service affects earnings differently across individuals
 - ▶ Health insurance has different effects on different patients
- With heterogeneous effects, **which** average treatment effect does IV estimate?
- Answer (Imbens & Angrist, 1994): the **Local Average Treatment Effect (LATE)** — the average effect *for the subpopulation whose treatment status is changed by the instrument*

Potential Outcomes Framework

- Each individual i has two **potential outcomes**:

$$\begin{aligned} Y_{1i} & \text{ outcome if treated } (D_i = 1) \\ Y_{0i} & \text{ outcome if not treated } (D_i = 0) \end{aligned}$$

- The **observed** outcome is:

$$Y_i = Y_{0i} + (Y_{1i} - Y_{0i})D_i$$

- The **individual treatment effect** is $\delta_i = Y_{1i} - Y_{0i}$

- Key estimands:

$$\text{ATE} = E[Y_{1i} - Y_{0i}]$$

$$\text{ATT} = E[Y_{1i} - Y_{0i} \mid D_i = 1]$$

$$\text{LATE} = E[Y_{1i} - Y_{0i} \mid \text{complier}] \quad (\text{defined below})$$

- The **fundamental problem of causal inference**: we never observe both Y_{1i} and Y_{0i} for the same person

- **Stable Unit Treatment Value Assumption (SUTVA)** requires:

1. **No interference:** individual i 's potential outcomes do not depend on the treatment status of others

$$Y_i = Y_i(D_i) \quad \text{not} \quad Y_i = Y_i(D_1, D_2, \dots, D_n)$$

2. **No hidden versions of treatment:** there is only one version of $D_i = 1$

- SUTVA rules out **general equilibrium** and **spillover** effects
- Example violations:
 - ▶ Herd immunity: your vaccination affects my health outcomes
 - ▶ Education and wages: if many people get degrees, the wage premium changes
- Under SUTVA the potential outcome notation is well-defined

Introducing the Instrument

- Let $Z_i \in \{0, 1\}$ be a binary instrument that **encourages** treatment but does not determine it
- Define **potential treatment** as a function of the instrument:

D_{1i} treatment status of i if $Z_i = 1$

D_{0i} treatment status of i if $Z_i = 0$

- The **observed** treatment is:

$$D_i = D_{0i} + (D_{1i} - D_{0i})Z_i$$

- Similarly define potential outcomes indexed by both treatment d and instrument z :

$$Y_{dzi} \quad d \in \{0, 1\}, \quad z \in \{0, 1\}$$

- The **exclusion restriction** collapses this to Y_{di} only — the instrument has no direct effect

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Four Compliance Types

- Based on (D_{1i}, D_{0i}) , every individual belongs to one of four groups:

Type	D_{1i}	D_{0i}	Behaviour
Complier	1	0	Treated iff $Z_i = 1$; responds to encouragement
Always-taker	1	1	Always treated regardless of Z_i
Never-taker	0	0	Never treated regardless of Z_i
Defier	0	1	Does the opposite of assignment

- We cannot observe compliance type — only (Y_i, D_i, Z_i) are observed
- Monotonicity** rules out defiers
- LATE is the average treatment effect **for compliers only**

LATE Assumption 1: Independence

- Independence (Instrument Exogeneity):

$$\{Y_{0i}, Y_{1i}, D_{0i}, D_{1i}\} \perp\!\!\!\perp Z_i$$

- The instrument is **as good as randomly assigned**
- Stronger than standard IV exogeneity: requires independence of all potential outcomes and potential treatments, not just zero correlation with the error term
- In practice it often comes from **randomization**:
 - ▶ Draft lottery numbers (Angrist, 1990): random assignment ensures $Z_i \perp (Y_{0i}, Y_{1i}, D_{0i}, D_{1i})$
 - ▶ Quarter of birth (Angrist & Krueger, 1991): arguably random conditional on covariates
 - ▶ Geographic distance to college (Card, 1995)
- When randomization holds only conditional on X_i : replace with conditional independence $(Y_{di}, D_{zi}) \perp Z_i \mid X_i$

LATE Assumption 2: Exclusion Restriction

- **Exclusion Restriction:**

$$Y_{dzi} = Y_{di} \quad \text{for } d = 0, 1 \text{ and all } z$$

- The instrument affects outcomes **only through** its effect on treatment D_i ; it has no direct effect on Y_i
- This collapses Y_{dzi} to Y_{di} , so observed outcomes can be written:

$$Y_i = Y_{0i} + (Y_{1i} - Y_{0i})D_i$$

- **Example:** in the draft lottery,
 - ▶ $Z_i = 1$ if lottery number was low (eligible for draft)
 - ▶ Exclusion: having a low lottery number affects earnings *only* through military service; lottery number itself has no direct effect on wages
- Often the most debatable assumption in practice — requires institutional knowledge and cannot be tested directly

LATE Assumption 3: First Stage

- First Stage (Relevance):

$$E[D_{1i} - D_{0i}] \neq 0 \iff E[D_i | Z_i = 1] \neq E[D_i | Z_i = 0]$$

- The instrument has a non-zero average effect on treatment — there must be a non-trivial proportion of **compliers**
- This is the standard IV relevance condition: $\text{Cov}(Z_i, D_i) \neq 0$
- In practice: tested with first-stage F -statistic
 - ▶ Rule of thumb: $F > 10$ (Staiger & Stock, 1997)
 - ▶ Weak first stage $\Rightarrow$ IV estimator is nearly undefined: small denominator amplifies any bias in the numerator
 - ▶ Weak instrument robust inference: Anderson-Rubin test, conditional likelihood ratio

LATE Assumption 4: Monotonicity

- Monotonicity (No Defiers):

$$D_{1i} \geq D_{0i} \quad \text{for all } i$$

- The instrument weakly **increases** everyone's probability of treatment — no one is discouraged from treatment by a higher Z_i
- Rules out **defiers**: individuals who do the opposite of what the instrument encourages
- This is a restriction on *heterogeneity in compliance*, not a restriction on unobservables
- Often plausible when Z_i is an encouragement or offer:
 - ▶ Being offered a voucher weakly increases program participation
 - ▶ Having a low draft number weakly increases military service
- Under monotonicity, the four types reduce to three: **compliers**, always-takers, and never-takers — defiers are ruled out

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The LATE Theorem (Imbens & Angrist, 1994)

Proposition (LATE Theorem)

Under independence, exclusion restriction, first stage, and monotonicity:

$$\frac{E[Y_i | Z_i = 1] - E[Y_i | Z_i = 0]}{E[D_i | Z_i = 1] - E[D_i | Z_i = 0]} = E[Y_{1i} - Y_{0i} | D_{1i} > D_{0i}]$$

- The left-hand side is the **Wald estimator** — the IV ratio with a binary instrument
- The right-hand side is the **Local Average Treatment Effect**: the average causal effect **for compliers**
- IV does **not** identify ATE or ATT in general — only LATE
- The word “local” means the estimate is local to the complier subpopulation *defined by this instrument* — it changes with the instrument

Step 1 — Reduced Form (Numerator):

$$E[Y_i | Z_i = 1] - E[Y_i | Z_i = 0]$$

By **independence** ($Z_i \perp (Y_{0i}, Y_{1i}, D_{0i}, D_{1i})$):

$$= E[Y_{0i} + (Y_{1i} - Y_{0i})D_{1i}] - E[Y_{0i} + (Y_{1i} - Y_{0i})D_{0i}]$$

By **exclusion restriction** ($Y_i = Y_{0i} + (Y_{1i} - Y_{0i})D_i$), this simplifies to:

$$= E[(Y_{1i} - Y_{0i})(D_{1i} - D_{0i})]$$

By **monotonicity** ($D_{1i} - D_{0i} \geq 0$) and the law of total expectation:

$$= E[Y_{1i} - Y_{0i} | D_{1i} > D_{0i}] \cdot P(D_{1i} > D_{0i})$$

Step 2 — First Stage (Denominator):

$$E[D_i | Z_i = 1] - E[D_i | Z_i = 0] \stackrel{\text{indep}}{=} E[D_{1i}] - E[D_{0i}] \stackrel{\text{mono}}{=} P(D_{1i} > D_{0i})$$

Step 3 — Taking the Ratio:

$$\frac{E[Y_i | Z_i = 1] - E[Y_i | Z_i = 0]}{E[D_i | Z_i = 1] - E[D_i | Z_i = 0]} = \frac{E[Y_{1i} - Y_{0i} | D_{1i} > D_{0i}] \cdot P(D_{1i} > D_{0i})}{P(D_{1i} > D_{0i})}$$
$$= E[Y_{1i} - Y_{0i} | D_{1i} > D_{0i}] \quad \blacksquare$$

Role of each assumption:

1. **Independence:** replaced $D_i | Z_i = z$ with D_{zi} , and similarly for Y_i
2. **Exclusion:** collapsed Y_{dzi} to Y_{di} ; removed the direct effect of Z on Y
3. **Monotonicity:** ensured $E[(Y_{1i} - Y_{0i})(D_{1i} - D_{0i})]$ only involves compliers; without it, the numerator would mix complier and defier effects, and the ratio would not be interpretable as a treatment effect

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What Population Does LATE Describe?

- LATE describes the treatment effect for **compliers** — those whose treatment status is switched by the instrument
- The complier share is identified from observables:

$$P(\text{complier}) = E[D_i \mid Z_i = 1] - E[D_i \mid Z_i = 0]$$

- What we **cannot** identify:
 - ▶ ATE: requires identifying effects for always-takers and never-takers too
 - ▶ ATT: requires identifying effects for always-takers too
- LATE = ATE iff treatment effects are **homogeneous** ($\delta_i = \delta$ for all i)
- LATE = ATT iff there are **no always-takers**: $P(D_{0i} = 1) = 0$
- LATE depends on the instrument: two valid instruments may give different LATE estimates because they define different complier populations

Estimand	Population	Typical estimator	Requires
ATE	All i	Experiment	Full randomisation
ATT	Treated i	DiD, matching	CIA or parallel trends
LATE	Compliers	IV/2SLS	4 LATE assumptions

- **External validity:** LATE may not generalise beyond the specific complier group defined by the instrument used
- **Policy relevance:** if the policy works through the same compliance mechanism as the instrument (e.g. a voucher program), LATE is the *right* estimand
- **Instrument comparison:** two valid instruments giving different 2SLS estimates are both correct — for different complier groups

Characterising Compliers

- We cannot identify individual compliers, but we can describe the **complier population** using observable covariates X_i
- The relative likelihood of characteristic x among compliers (Angrist & Pischke, MHE eq. 4.4.9):

$$\frac{P(X_i = x \mid \text{complier})}{P(X_i = x)} = \frac{E[D_{1i} - D_{0i} \mid X_i = x]}{E[D_{1i} - D_{0i}]}$$

- Estimated by running the first stage separately for $X_i = x$:

$$\frac{E[D_i \mid Z_i = 1, X_i = x] - E[D_i \mid Z_i = 0, X_i = x]}{E[D_i \mid Z_i = 1] - E[D_i \mid Z_i = 0]}$$

- Ratios > 1 : characteristic x over-represented among compliers
- This allows researchers to say *who* the compliers are, which is essential for judging external validity (Abadie, 2003)

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From Wald Estimator to 2SLS

- With a binary instrument and no covariates:

$$\hat{\delta}^{IV} = \frac{\overline{Y}_{Z=1} - \overline{Y}_{Z=0}}{\overline{D}_{Z=1} - \overline{D}_{Z=0}}$$

- With covariates X_i : 2SLS estimates a **covariate-weighted average** of conditional LATEs, weighting by the complier share in each cell
- With multiple instruments $Z_{1i}, \dots, Z_{ji}$: 2SLS estimates a **precision-weighted average** of instrument-specific LATEs:

$$\hat{\delta}^{2SLS} \approx \sum_{k=1}^j \omega_k \cdot \text{LATE}_k$$

where $\omega_k \propto$ first-stage F -statistic of instrument k

- Instruments with stronger first stages receive higher weight — they identify LATE for their compliers more precisely

IV with Covariates and Conditional LATE

- When independence holds only conditional on covariates X_i :

$$\{Y_{0i}, Y_{1i}, D_{0i}, D_{1i}\} \perp\!\!\!\perp Z_i \mid X_i$$

- The 2SLS estimator identifies a weighted average of **conditional LATEs**:

$$\delta^{LATE} = \sum_x E[Y_{1i} - Y_{0i} \mid D_{1i} > D_{0i}, X_i = x] \cdot \omega(x)$$

where $\omega(x) \propto P(D_{1i} > D_{0i} \mid X_i = x) \cdot P(X_i = x)$

- Practical implementation:**

1. First stage: regress D_i on Z_i and X_i , obtain $\hat{D}_i$
2. Second stage: regress Y_i on $\hat{D}_i$ and X_i
3. Compute SE using **structural residuals** $\hat{e}_i = Y_i - X_i' \hat{\gamma} - D_i \hat{\delta}$, not second-stage residuals

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Application 1: Draft Lottery and Earnings (Angrist, 1990)

- **Question:** What is the causal effect of military service on earnings?
- **Problem:** Selection bias — those who serve may differ systematically from non-servers
- **Instrument:** Vietnam-era draft lottery number ($Z_i \in \{1, \dots, 365\}$, randomly assigned)
 - ▶ *Independence:* lottery numbers randomly assigned $\Rightarrow$ exogenous
 - ▶ *Exclusion:* lottery number affects earnings only through military service
 - ▶ *Monotonicity:* low lottery number weakly increases probability of service
- **Compliance types:**
 - ▶ *Compliers:* serve only because of low lottery number (draft-induced)
 - ▶ *Always-takers:* volunteers — serve regardless of lottery number
 - ▶ *Never-takers:* avoid service regardless (deferments, health exemptions)
- **LATE:** effect of military service on earnings for *draft compliers*

Application 1: Draft Lottery (cont'd)

- **Wald estimator** (low number = $Z_i = 1$, high number = $Z_i = 0$):

$$\hat{\delta}^{LATE} = \frac{E[\text{Earnings}_i \mid Z_i = 1] - E[\text{Earnings}_i \mid Z_i = 0]}{E[D_i \mid Z_i = 1] - E[D_i \mid Z_i = 0]}$$

- **Results:** White veterans earned $\approx 15\%$ less than non-veterans; effect for non-whites smaller in magnitude
- **LATE interpretation:** this is the earnings effect for *draft compliers* — men who served because of their lottery number but would not have volunteered
- **External validity:** does this estimate apply to volunteers (always-takers)?
 - ▶ Not necessarily — volunteers self-select for reasons potentially correlated with earnings
 - ▶ LATE makes this limitation *explicit* and forces the researcher to be clear about which population is identified

Application 2: Compulsory Schooling (Angrist & Krueger, 1991)

- **Question:** What is the causal return to an additional year of schooling?
- **Problem:** ability bias — more able individuals get more schooling *and* earn more
- **Instrument:** quarter of birth Z_i
 - ▶ US compulsory schooling: must stay until age 16; school entry cutoff Jan 1
 - ▶ Q1 births enter school oldest $\Rightarrow$ can drop out with **less** schooling than Q4 births
 - ▶ Quarter of birth is (arguably) randomly determined relative to ability
- **Compliance types:**
 - ▶ *Compliers*: marginal students who drop out at 16 — constrained by the compulsory schooling law
 - ▶ *Always-takers*: complete more schooling regardless of birth quarter
 - ▶ *Never-takers*: drop out regardless of birth quarter
- **LATE:** return to schooling for *compulsory-schooling compliers* — those at the margin of dropping out

Application 2: Compulsory Schooling (cont'd)

- **Structural equation:**

$$\log(w_i) = \mu + \rho s_i + X_i' \gamma + \varepsilon_i$$

- **First stage:** $s_i = X_i' \pi + \sum_j \alpha_j Z_{ij} + \eta_i$, where Z_{ij} are quarter-of-birth dummies
- **Exclusion restriction:** quarter of birth affects wages only through years of schooling
 - ▶ Contested: Q1 births may differ in school maturity, parental age, season-of-birth health effects
- **Results:** return to schooling $\approx 7\text{--}10\%$ per year
- **LATE interpretation:** this is the return for *marginal students* bound by compulsory schooling
 - ▶ These are students with the *lowest* desired schooling — potentially lower ability
 - ▶ Diminishing returns would suggest $\text{LATE} < \text{ATE}$ for these students — yet estimates are similar to OLS, suggesting ability bias in OLS may be modest

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LATE with a Continuous Instrument

- With a continuous instrument Z_i (e.g. distance to college, alcohol taxes), IV still has a LATE interpretation
- **Angrist & Imbens (1995)**: with a multi-valued instrument, 2SLS estimates a weighted average of **pairwise LATEs**:

$$\delta^{IV} = \sum_{z > z'} w_{zz'} \cdot E[Y_{1i} - Y_{0i} \mid D_{zi} > D_{z'i}]$$

- **Marginal Treatment Effect (MTE)** provides the most general framework (Heckman & Vytlacil, 2005):

$$\text{MTE}(u) = E[Y_{1i} - Y_{0i} \mid U_i = u]$$

where U_i is resistance to treatment

- ATE, ATT, LATE, and 2SLS can all be written as:

$$\delta^* = \int_0^1 \text{MTE}(u) \cdot \omega^*(u) du$$

with different weight functions ω^* — this unifies all estimands

Partial Compliance and Fuzzy RDD

- The LATE framework applies beyond binary instruments — it is the foundation for **fuzzy regression discontinuity designs (fuzzy RDD)**
- In a fuzzy RDD:

$$Z_i = \mathbf{1}[X_i \geq c] \quad (\text{instrument: being above the threshold})$$

- ▶ Treatment probability jumps discontinuously at threshold c but does not jump to 1
 - ▶ Not everyone above the threshold takes treatment (partial compliance)
- The fuzzy RDD estimator is a Wald ratio at the threshold:

$$\hat{\theta}^{RDD} = \frac{\lim_{x \downarrow c} E[Y_i | X_i = x] - \lim_{x \uparrow c} E[Y_i | X_i = x]}{\lim_{x \downarrow c} E[D_i | X_i = x] - \lim_{x \uparrow c} E[D_i | X_i = x]}$$

- Under the LATE assumptions (with independence replaced by local continuity), this identifies LATE **at the threshold** for local compliers

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- With **heterogeneous treatment effects**, IV does not estimate ATE or ATT
- Under four assumptions (independence, exclusion, first stage, monotonicity), IV identifies the **Local Average Treatment Effect**:

$$\frac{E[Y_i|Z_i = 1] - E[Y_i|Z_i = 0]}{E[D_i|Z_i = 1] - E[D_i|Z_i = 0]} = E[Y_{1i} - Y_{0i} \mid D_{1i} > D_{0i}]$$

- Key takeaways:
 - ▶ LATE is the average effect **for compliers** — those switched by the instrument
 - ▶ Different instruments identify LATE for *different* complier populations
 - ▶ LATE raises external validity concerns but is often the policy-relevant estimand
 - ▶ 2SLS with multiple instruments gives a precision-weighted average of LATEs
 - ▶ Complier characteristics can be partially identified from the data
 - ▶ LATE is also the foundation for fuzzy RDD
- **Reference:** Angrist & Pischke, *Mostly Harmless Econometrics*, Chapter 4